Host Firewall Lab with UFW on Ubuntu

Controlling access to a web server with UFW (Uncomplicated Firewall)

Author: Muhammad Aun Hasan

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Platform: VMware Workstation 17.5 · Ubuntu 22.04 virtual machine · Windows 11 host

Based on: the Firewall Lab lesson in TCM Security's *Practical Security Fundamentals* course

Summary

I built a small web server inside an Ubuntu virtual machine and used **UFW (Uncomplicated Firewall)** to control who can reach it. I tested every rule from my Windows computer on the same home network. The lab shows how a firewall blocks traffic by default, how to open one port, how to block one computer, and why **the order of rules matters**.

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Lab at a glance

Item	Value
Hypervisor (virtual machine software)	VMware Workstation 17.5
Virtual machine	Ubuntu 22.04, 4 GB memory, 2 processors, 30 GB disk
Virtual machine name / user	`smahn-virtual-machine` / `smahn`
Network mode	Bridged (changed from NAT)
Ubuntu IP address	`192.168.18.54/24` (interface `ens33`)
Windows host IP address	`192.168.18.26`
Home router (gateway)	`192.168.18.1`
Web server	Python `http.server` on port **8080** (TCP)
Web page	`~/firewall_lab/index.html`
Firewall	UFW (Uncomplicated Firewall)

Diagram



Both computers are on the home network behind the router 192.168.18.1.

Skills shown

- Setting up virtual machine networking (NAT vs Bridged)
- Linux command line (`mkdir` , `apt` , `ip a` , `python3`)
- Running a simple web server for testing
- Host firewall setup with UFW: enable, allow, deny, delete, numbered rules
- Understanding default-deny and **first-match rule order**
- Testing network access from a second computer

Part A - Prepare the virtual machine

Step 1. The virtual machine's network adapter was on NAT, which hides the virtual machine behind the host.

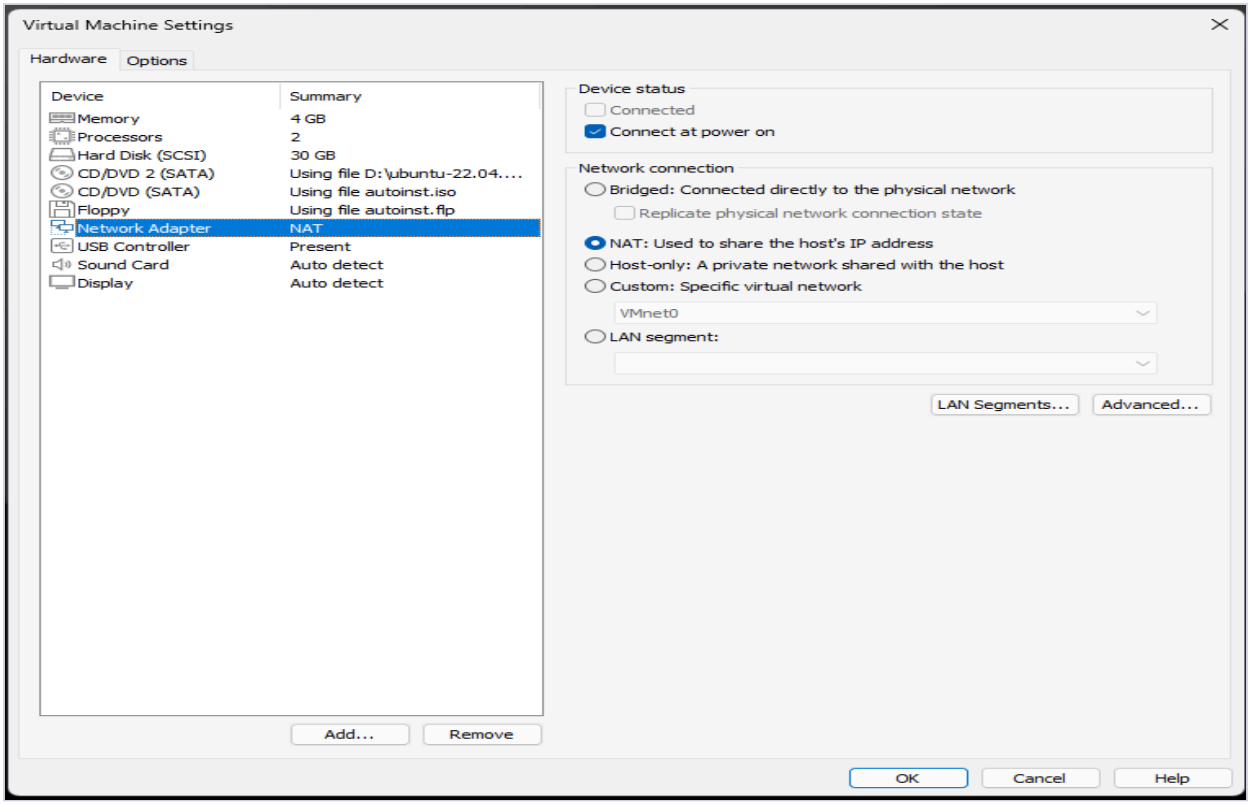


Figure 1 - Network adapter on NAT

Step 2. I changed it to **Bridged**, so the virtual machine gets its own address on my home network and Windows can reach it directly.

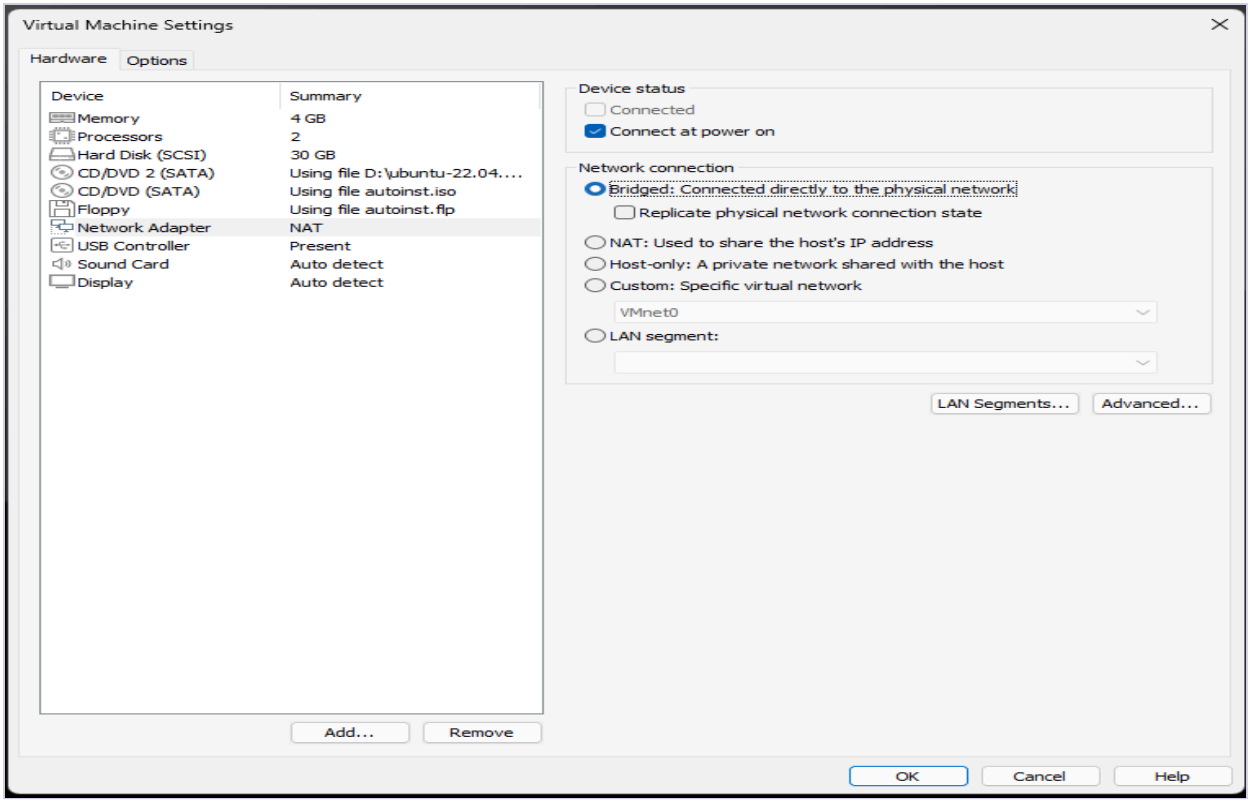


Figure 2 - Network adapter changed to Bridged

Part B - Build a test web server

Step 3. I made a folder for the lab and moved into it.

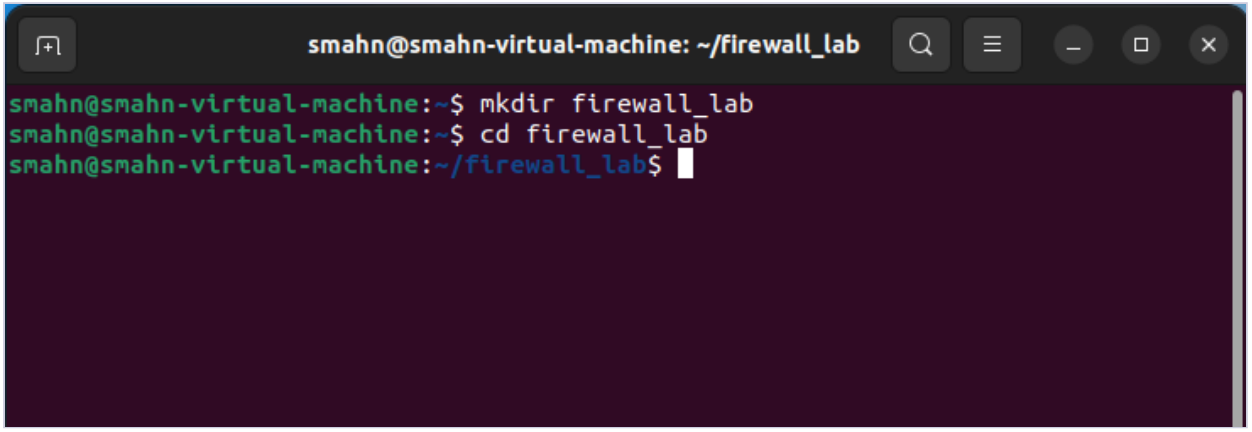


Figure 3 - Creating the firewall_lab folder

Step 4. I made sure the gedit text editor was installed, then used it to write `index.html` with one line: "*I'm able to see this through the firewall webpage*".

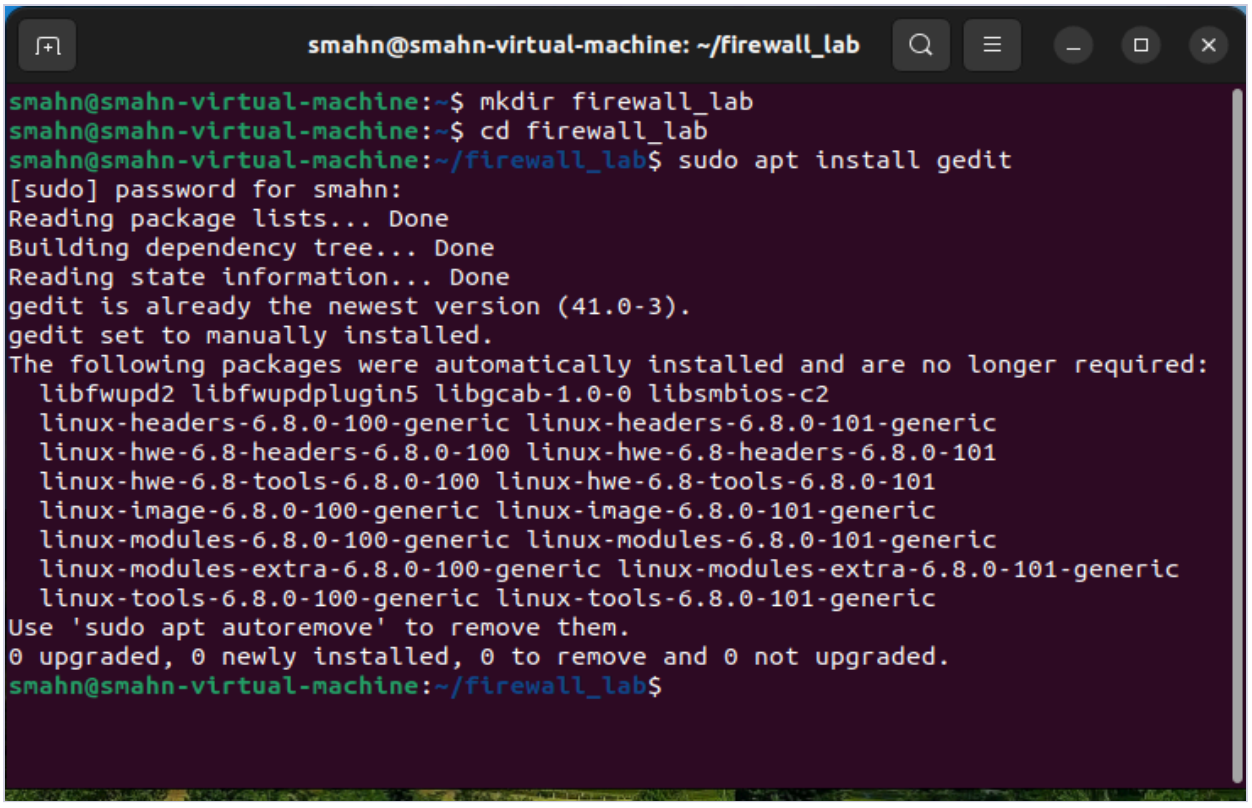


Figure 4 - Installing gedit

Step 5. I started a web server on port 8080 with `python3 -m http.server 8080` and opened `http://127.0.0.1:8080` inside Ubuntu. The page loaded (`GET / 200`).

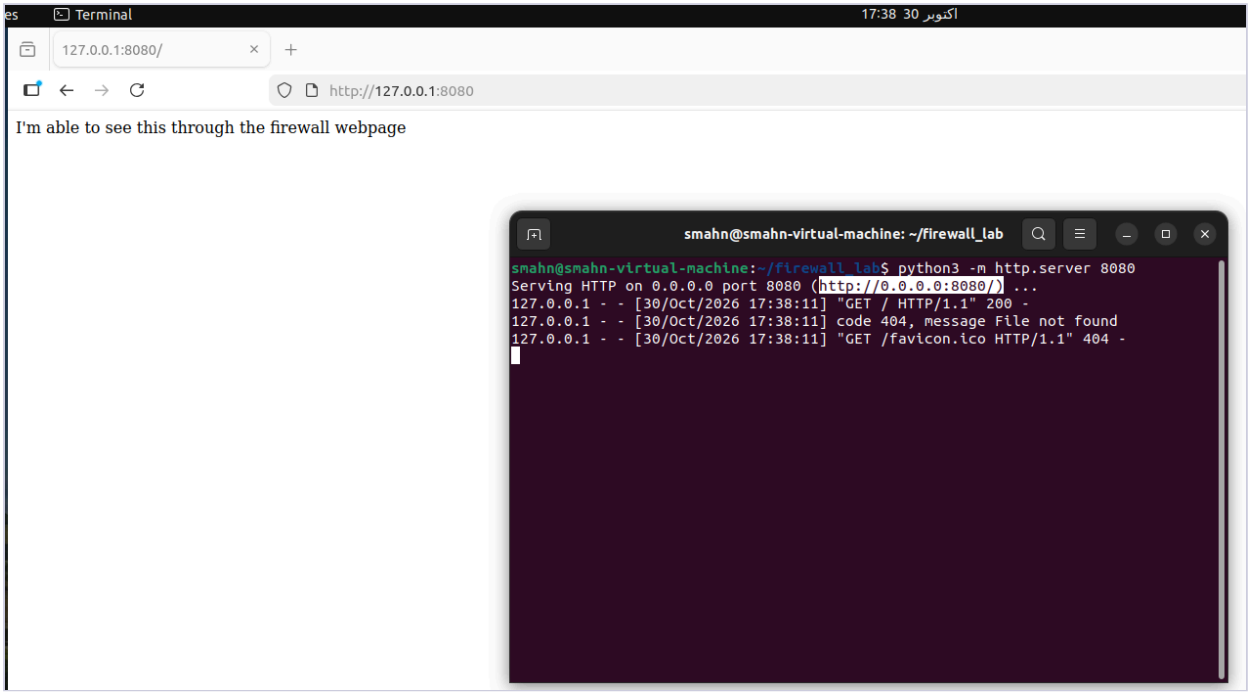


Figure 5 - Web page works locally

Step 6. I checked the firewall. UFW was **inactive** (off).

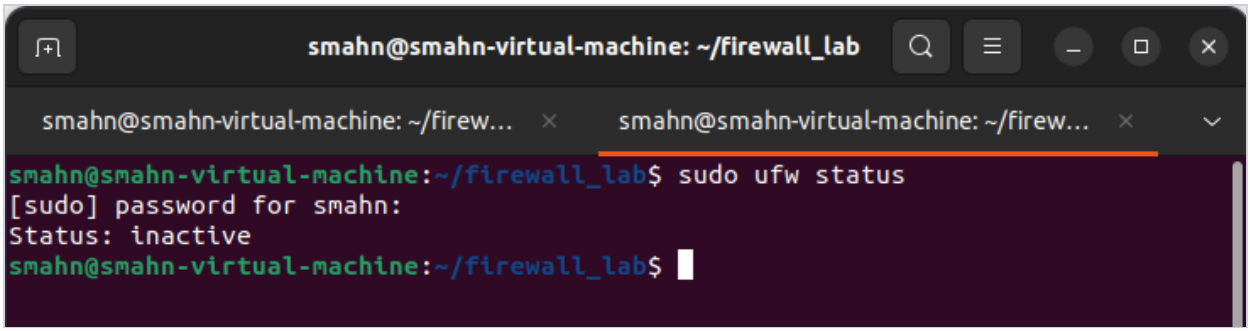


Figure 6 - UFW is off

Step 7. `ip a` showed the Ubuntu address: **192.168.18.54**.

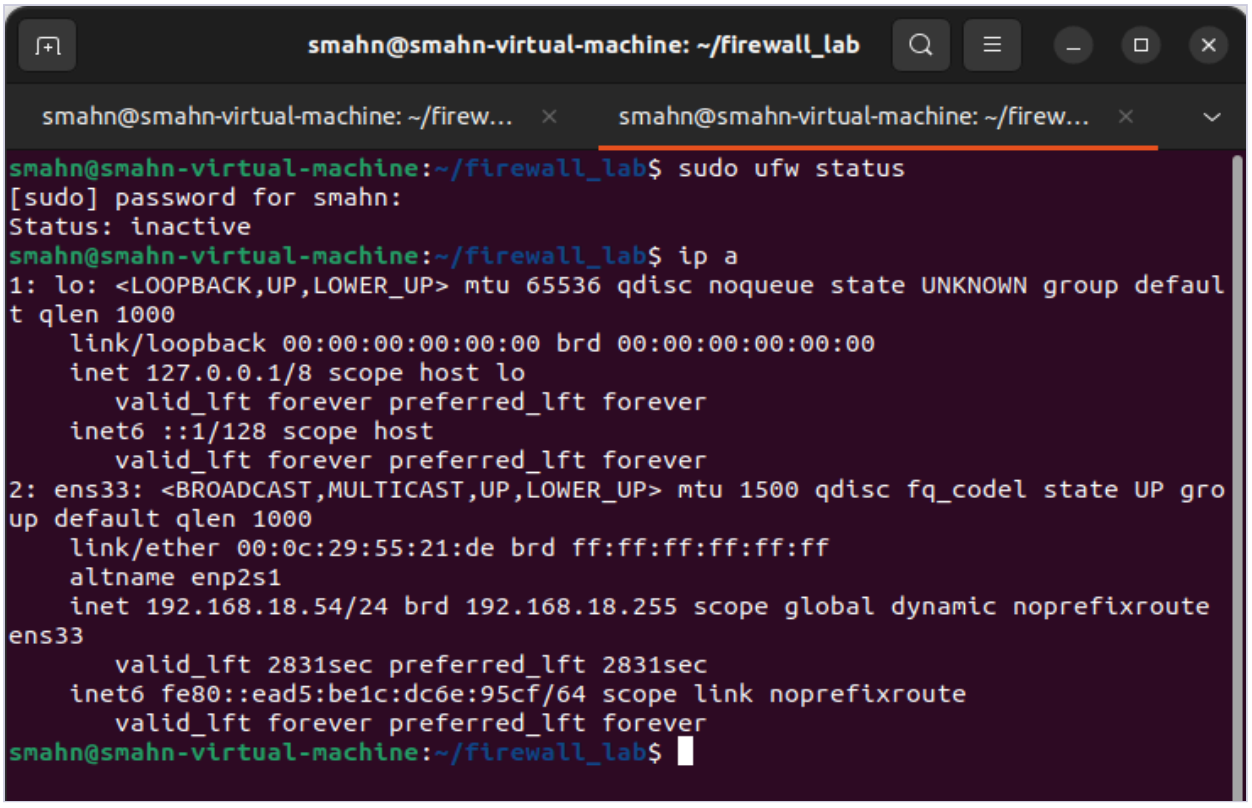


Figure 7 - Ubuntu IP address

Step 8. `ipconfig` on Windows showed the host address: **192.168.18.26** (public IPv6 addresses blurred).

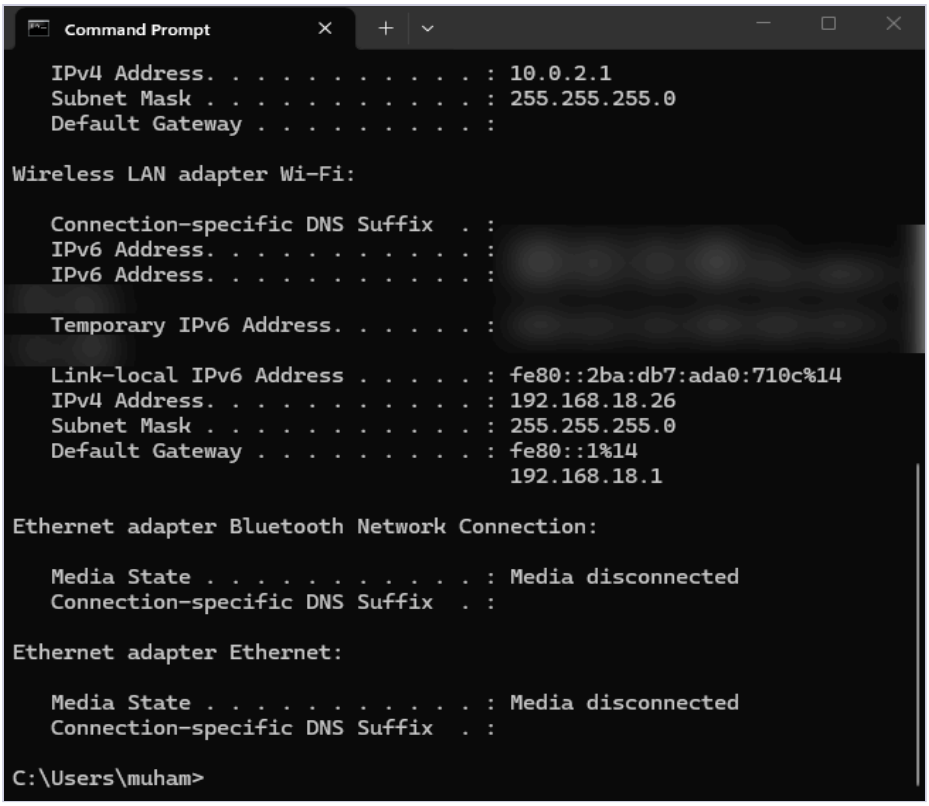


Figure 8 - Windows host IP address

Step 9. From Windows, I opened `http://192.168.18.54:8080`. With the firewall off, the page loaded.

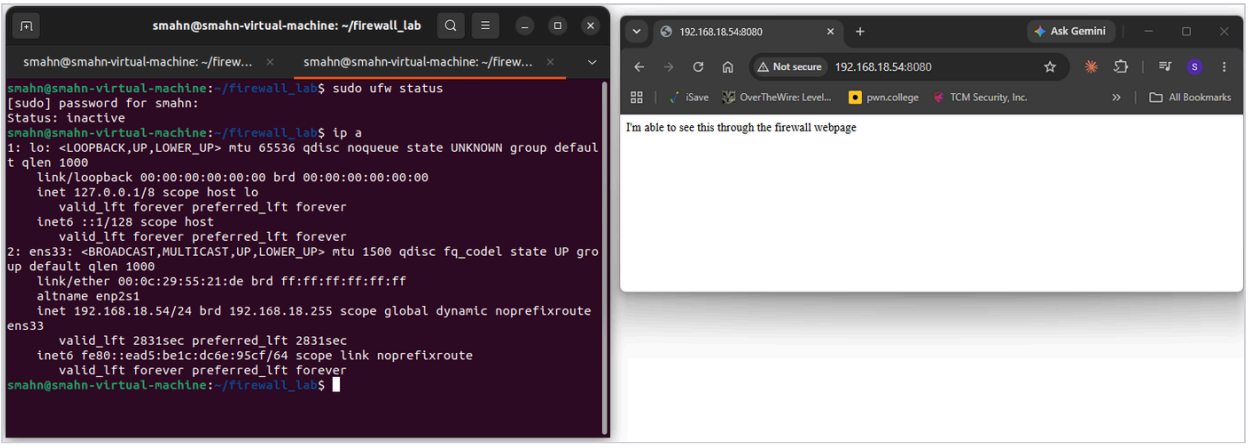


Figure 9 - Windows reaches the server (firewall off)

Part C - Turn on the firewall

Step 10. I turned the firewall on with `sudo ufw enable`. Status is now **active**.

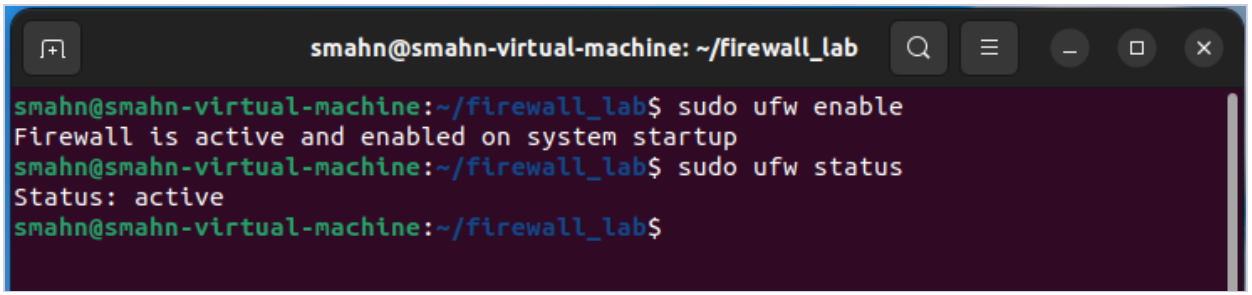


Figure 10 - UFW turned on

Step 11. I started the server again and reloaded the page from Windows. It failed with "*This site can't be reached*", because UFW blocks all incoming traffic by default.

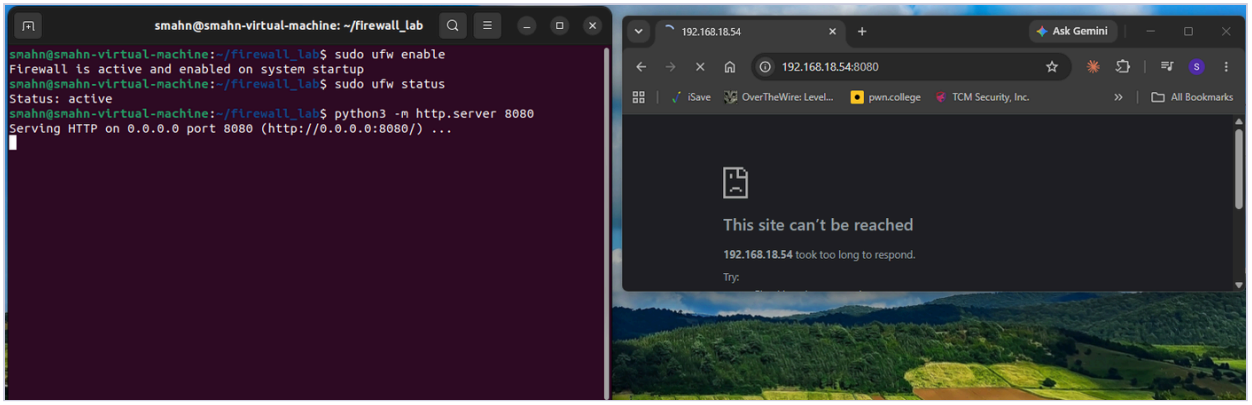


Figure 11 - Blocked by the default rule

Step 12. I opened the port with `sudo ufw allow 8080/tcp` (rule added for IPv4 and IPv6). The page loaded from Windows again.

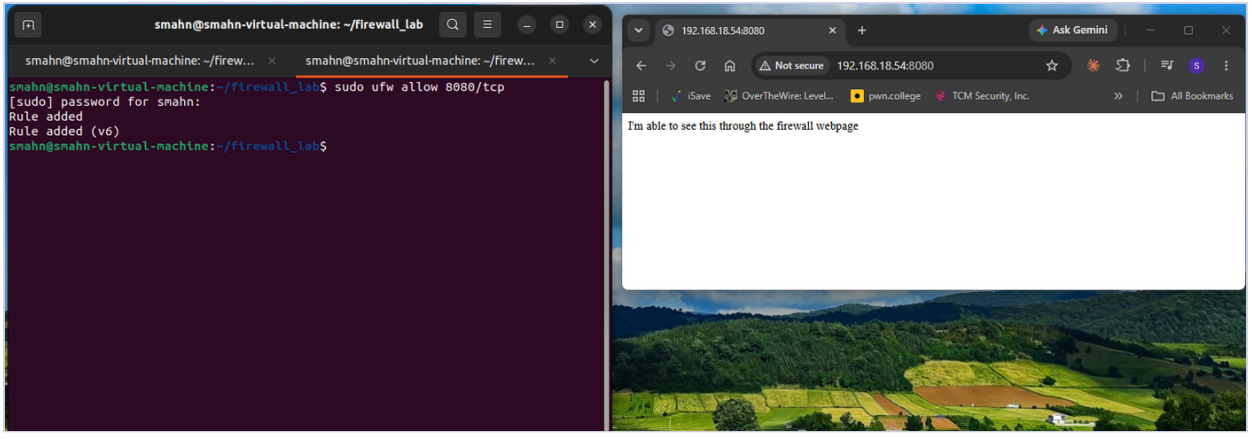


Figure 12 - Port 8080 allowed, page loads

Part D - Block one computer and test rule order

Step 13. I added a rule to block only my Windows computer on port 8080: `sudo ufw deny from 192.168.18.26 to any port 8080 proto tcp`

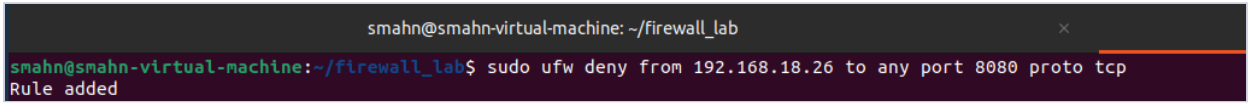


Figure 13 - Deny rule for the Windows host

Step 14. The deny rule was placed **below** the allow rule. UFW uses the **first rule that matches**, so the allow rule won and the page **still loaded**.

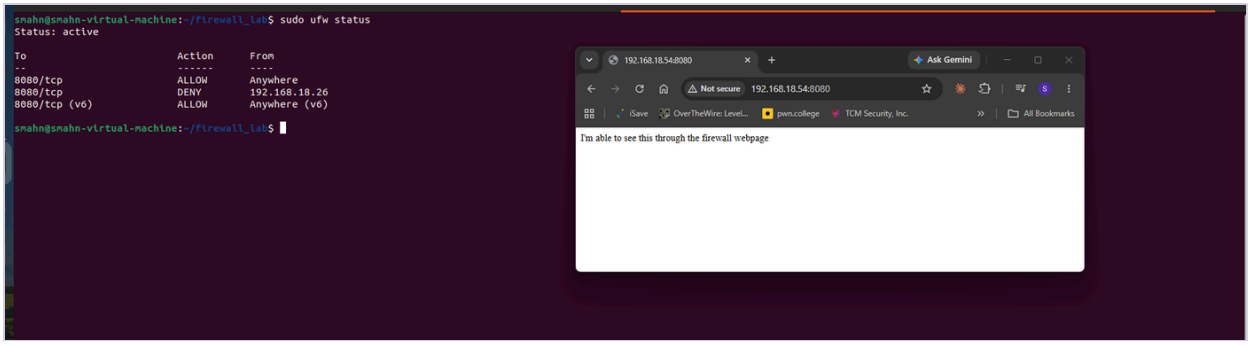


Figure 14 - Allow rule is first, so the deny rule is never reached

Step 15. I listed the rules with numbers (`sudo ufw status numbered`) and removed the allow rule with `sudo ufw delete 1`. Now the deny rule was first, and Windows was **blocked**.

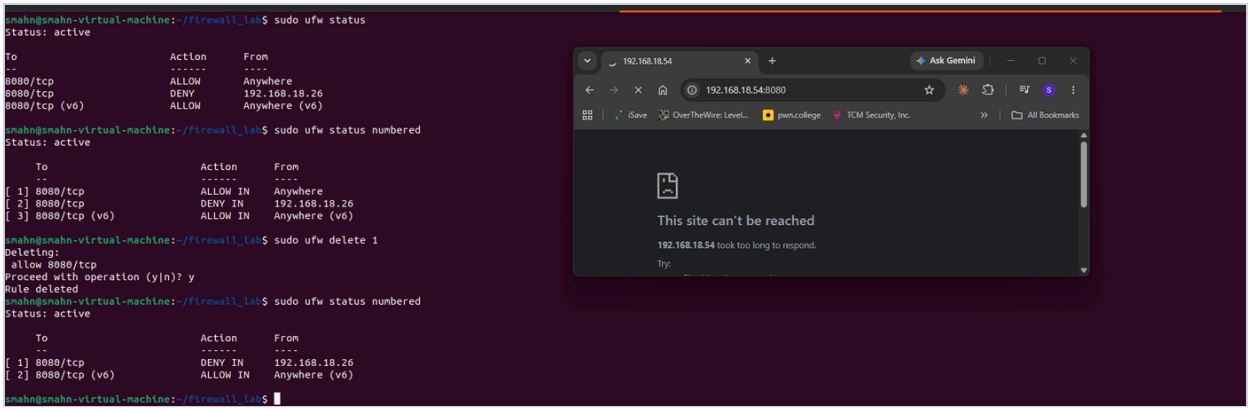


Figure 15 - Allow rule deleted, Windows host blocked

Step 16. I added `sudo ufw allow 8080/tcp` again. It went **below** the deny rule, so other computers can reach port 8080 but my Windows computer stays blocked. I tested it from Windows and the page was blocked.

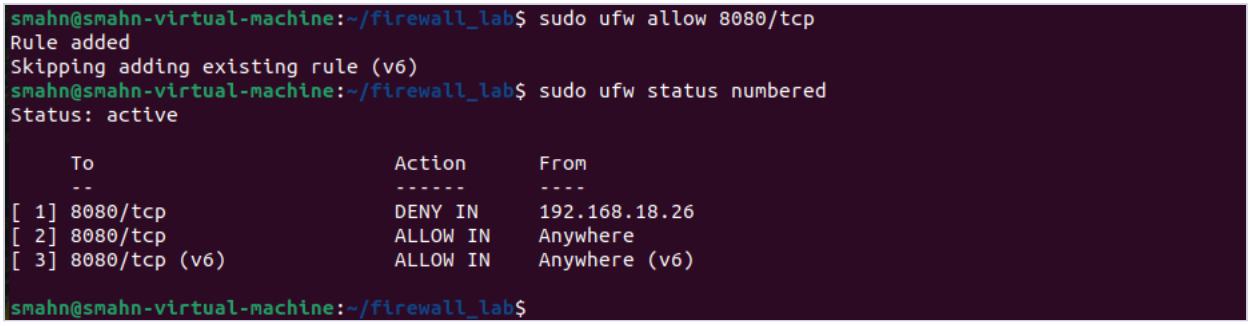


Figure 16 - Final rules: deny the Windows host first, then allow everyone else

Final rule list

#	To	Action	From
1	8080/tcp	DENY IN	192.168.18.26
2	8080/tcp	ALLOW IN	Anywhere
3	8080/tcp (v6)	ALLOW IN	Anywhere (v6)

Test results

All tests were made from the Windows host (`192.168.18.26`) to `http://192.168.18.54:8080`.

Test	Expected	Result
Page inside Ubuntu (`127.0.0.1:8080`)	Loads	✅ Loaded
From Windows, firewall off	Loads	✅ Loaded
From Windows, firewall on, no rules	Blocked	✅ Blocked
From Windows, `allow 8080/tcp`	Loads	✅ Loaded
Deny Windows host added **below** allow	Still loads (allow matches first)	✅ Loaded
Allow rule deleted, only deny left	Blocked	✅ Blocked
Allow re-added **below** deny	Windows host blocked	✅ Blocked

Glossary

Short form	Meaning
UFW	Uncomplicated Firewall - the easy firewall tool on Ubuntu
VM	Virtual Machine - a computer running inside another computer
NAT	Network Address Translation - the virtual machine shares the host's address and is hidden behind it
Bridged	The virtual machine joins the home network as its own device
IP	Internet Protocol - the address of a device on a network
IPv4 / IPv6	Version 4 / version 6 of IP addresses (`v6` in UFW means IPv6)
TCP	Transmission Control Protocol - the connection type web pages use
HTTP	HyperText Transfer Protocol - how web pages are sent
Port 8080	The "door" number the test web server listens on
`/24`	Subnet size - the first three parts of the address are the same network (255.255.255.0)
`sudo`	Run a command as administrator on Linux
`apt`	Ubuntu's tool to install software
`ip a`	Linux command that shows network addresses
`ipconfig`	Windows command that shows network addresses
127.0.0.1	"This computer" (loopback address)

Disclaimer

This lab was built for learning, in my own home lab and on my own network. It is not a production setup. I will further tighten security controls as I keep building on it.

Repository layout

```
Firewall-Lab/
├── README.md
├── Firewall-Lab.pdf
└── images/
    ├── 01-vm-network-nat.png
    ├── ...
    └── 16-re-add-allow-deny-first.png
```